

# OpenStack Complete Command Reference

## Basic • Advanced • Troubleshooting — With Explanations

All commands use the **unified** `openstack` CLI (python-openstackclient).

Run `source admin-openrc.sh` (or your keystone rc file) before using any command.

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## 1. Authentication & Environment

## Load Credentials

```
# Source the OpenRC file to set environment variables
source ~/admin-openrc.sh

# Verify you are authenticated – prints your current username
openstack token issue

# Show all currently exported OpenStack environment variables
env | grep OS_

# Check which project/tenant you are working in
openstack project show $(openstack token issue -f value -c project_id)
```

**Why:** Every OpenStack CLI command needs credentials. The `openrc` file exports `OS_AUTH_URL`, `OS_USERNAME`, `OS_PASSWORD`, `OS_PROJECT_NAME`, etc. Without sourcing it, all commands return "Missing value auth-url".

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## Switch Between Projects

```
# List all projects you have access to
openstack project list

# Switch to a different project without editing openrc
export OS_PROJECT_NAME=myproject

# Create a scoped token for a specific project
openstack token issue --os-project-name myproject
```

---

## Application Credentials (Token-less Auth)

```
# Create application credentials (for scripts / CI – no password exposed)
openstack application credential create my-app-cred \
  --role member \
  --expiration "2025-12-31T00:00:00"

# List application credentials
openstack application credential list

# Delete an application credential
openstack application credential delete my-app-cred
```

---

## 2. Identity – Keystone

### Users

```
# List all users
openstack user list
# Shows: ID, Name, Enabled status

# Show detailed info about a specific user
openstack user show admin

# Create a new user
openstack user create \
    --domain default \
    --password MyPass123! \
    --email user@example.com \
    john
# --domain: which Keystone domain (use "default" for single-domain setups)

# Update a user's password
openstack user set --password NewPass456! john

# Disable a user (they cannot log in but account is preserved)
openstack user set --disable john

# Re-enable a user
openstack user set --enable john

# Delete a user
openstack user delete john

# List users in a specific domain
openstack user list --domain default
```

---

### Projects (Tenants)

```
# List all projects
openstack project list

# Show details of a project
```

```
openstack project show myproject

# Create a new project
openstack project create \
  --domain default \
  --description "Production workloads" \
  production

# Disable a project (prevents logins/resource creation)
openstack project set --disable production

# Delete a project
openstack project delete production
```

---

## Roles & Role Assignments

```
# List all available roles
openstack role list
# Common roles: admin, member, reader

# Create a custom role
openstack role create network-operator

# Assign a role to a user on a project
openstack role add \
  --project production \
  --user john \
  member
# This allows "john" to create resources in "production"

# Show all role assignments for a user
openstack role assignment list \
  --user john \
  --names

# Remove a role from a user
openstack role remove \
  --project production \
  --user john \
  member
```

---

## Domains & Groups

```
# List domains
openstack domain list

# Create a domain
openstack domain create engineering

# Create a group
openstack group create devteam --domain default

# Add user to a group
openstack group add user devteam john

# Assign a role to a whole group on a project
openstack role add \
    --project production \
    --group devteam \
    member
```

---

## Service Catalog & Endpoints

```
# List all registered services
openstack service list
# Shows: ID, Name, Type (identity, compute, image, etc.)

# Show details of a service
openstack service show nova

# List all endpoints
openstack endpoint list
# Shows: ID, Region, Service, Interface (public/internal/admin), URL

# List endpoints for a specific service
openstack endpoint list --service nova

# Create a new endpoint (rarely needed manually)
openstack endpoint create \
    --region RegionOne \
    compute public http://192.168.1.10:8774/v2.1

# Delete an endpoint
openstack endpoint delete <endpoint-id>
```

---

## 3. Compute – Nova

### Instances (Servers)

```
# List all instances in current project
openstack server list

# List all instances across ALL projects (admin only)
openstack server list --all-projects

# List with more details (IP, host, image)
openstack server list --long

# Show full details of a specific instance
openstack server show my-vm

# Shows: status, addresses, flavor, image, host, security groups, etc.

# Filter instances by status
openstack server list --status ERROR
openstack server list --status ACTIVE
openstack server list --status SHUTOFF
```

---

### Create Instances

```
# Basic instance creation
openstack server create \
  --image cirros \
  --flavor m1.small \
  --network private-net \
  my-vm

# Create with SSH keypair
openstack server create \
  --image ubuntu-22.04 \
  --flavor m1.medium \
  --network private-net \
  --key-name mykey \
  my-ubuntu-vm

# Create with user-data (cloud-init script)
openstack server create \
  --image ubuntu-22.04 \
  --flavor m1.medium \
```

```
--network private-net \  
--user-data ./cloud-init.sh \  
my-vm-with-init  
  
# Create with security group  
openstack server create \  
  --image cirros \  
  --flavor m1.tiny \  
  --network private-net \  
  --security-group web-sg \  
  web-server  
  
# Create with multiple NICs (two networks)  
openstack server create \  
  --image ubuntu-22.04 \  
  --flavor m1.medium \  
  --nic net-id=<net1-id> \  
  --nic net-id=<net2-id> \  
  dual-homed-vm  
  
# Boot from volume (root disk as Cinder volume)  
openstack server create \  
  --image ubuntu-22.04 \  
  --flavor m1.medium \  
  --boot-from-volume 50 \  
  --network private-net \  
  persistent-vm  
# --boot-from-volume 50 creates a 50GB Cinder root volume  
  
# Create multiple identical instances  
openstack server create \  
  --image cirros \  
  --flavor m1.tiny \  
  --network private-net \  
  --min 3 --max 3 \  
  web-server  
# Creates web-server-1, web-server-2, web-server-3
```

---

## Instance Lifecycle

```
# Stop (graceful shutdown via ACPI signal)  
openstack server stop my-vm  
  
# Start a stopped instance
```

```
openstack server start my-vm

# Reboot (soft = graceful OS reboot, hard = power cycle)
openstack server reboot my-vm          # soft reboot
openstack server reboot --hard my-vm    # hard reboot (like pulling plug)

# Suspend (freezes VM state to RAM – fast resume)
openstack server suspend my-vm

# Resume a suspended instance
openstack server resume my-vm

# Pause (freezes in hypervisor memory – instantaneous)
openstack server pause my-vm
openstack server unpause my-vm

# Shelve (saves to image, frees compute resources)
openstack server shelve my-vm

# Unshelve (restores from shelved state)
openstack server unshelve my-vm

# Delete an instance
openstack server delete my-vm

# Force delete (bypasses soft-delete)
openstack server delete --wait my-vm
```

---

## Console & Logs

```
# Get VNC console URL (open in browser)
openstack console url show my-vm
openstack console url show --novnc my-vm

# Get serial console (text output)
openstack console url show --serial my-vm

# View console log (last 50 lines of boot output)
openstack console log show my-vm

# View more lines
openstack console log show --lines 200 my-vm

# Very useful when SSH is unavailable – shows kernel boot messages, cloud-init
```

---

## Flavors

```
# List all flavors
openstack flavor list

# List including private flavors
openstack flavor list --all

# Show flavor details
openstack flavor show m1.large

# Create a new flavor
openstack flavor create \
    --ram 4096 \
    --vcpus 2 \
    --disk 40 \
    --public \
    m1.custom
# --ram in MB, --disk in GB

# Create a private flavor (only accessible to specific projects)
openstack flavor create \
    --ram 32768 \
    --vcpus 16 \
    --disk 200 \
    --private \
    gpu-large

# Grant access to private flavor for a project
openstack flavor set \
    --project myproject \
    gpu-large

# Add extra specs (e.g., CPU pinning, NUMA topology, GPU)
openstack flavor set \
    --property hw:cpu_policy=dedicated \
    --property hw:mem_page_size=1GB \
    m1.custom

# Delete a flavor
openstack flavor delete m1.custom
```

---

## Keypairs

```
# List keypairs
openstack keypair list

# Create a keypair (OpenStack generates the private key)
openstack keypair create mykey > mykey.pem
chmod 600 mykey.pem

# Import your existing public key
openstack keypair create \
  --public-key ~/.ssh/id_rsa.pub \
  mykey

# Show keypair fingerprint
openstack keypair show mykey

# Delete a keypair
openstack keypair delete mykey
```

---

## Server Actions (Resize, Migrate, Snapshot)

```
# Resize instance to a larger flavor
openstack server resize \
  --flavor m1.large \
  my-vm
# Instance goes to VERIFY_RESIZE status

# Confirm the resize (makes it permanent)
openstack server resize confirm my-vm

# Revert resize (go back to original flavor)
openstack server resize revert my-vm

# Live migrate (move VM to different host while running – admin only)
openstack server migrate \
  --live-migration \
  --host compute02.example.com \
  my-vm

# Cold migrate (instance must be stopped)
openstack server migrate my-vm

# Create a snapshot of a running instance
```

```
openstack server image create \  
    --name my-vm-snapshot \  
    my-vm  
# Creates a Glance image from the running instance  
  
# Rebuild instance (replace OS disk from image, keep same IP/ID)  
openstack server rebuild \  
    --image ubuntu-22.04 \  
    my-vm  
  
# Evacuate (move instance off a failed compute host – admin only)  
openstack server evacuate \  
    --host new-compute-host \  
    my-vm
```

---

## Compute Host & Hypervisor Info

```
# List all compute services and their states  
openstack compute service list  
# Shows: Binary, Host, Zone, Status (enabled/disabled), State (up/down)  
  
# Disable a compute node (for maintenance)  
openstack compute service set \  
    --disable \  
    --disable-reason "Hardware maintenance" \  
    compute01.example.com nova-compute  
  
# Re-enable a compute node  
openstack compute service set \  
    --enable \  
    compute01.example.com nova-compute  
  
# List hypervisors with resource usage  
openstack hypervisor list  
openstack hypervisor list --long  
  
# Show detailed hypervisor stats  
openstack hypervisor show compute01.example.com  
  
# Show hypervisor uptime  
openstack hypervisor show --wrap 120 compute01.example.com  
  
# List availability zones  
openstack availability zone list
```

```
openstack availability zone list --long
```

---

## Server Groups (Affinity/Anti-affinity)

```
# Create server group with anti-affinity (instances spread across hosts)
openstack server group create \
    --policy anti-affinity \
    web-tier-group
# Ensures no two web servers land on the same compute host

# Create server group with affinity (instances co-located on same host)
openstack server group create \
    --policy affinity \
    db-cluster-group

# Launch instance in a server group
openstack server create \
    --image ubuntu-22.04 \
    --flavor m1.medium \
    --network private-net \
    --hint group=<server-group-id> \
    web-server-1

# List server groups
openstack server group list

# Show server group details
openstack server group show web-tier-group
```

---

## 4. Networking – Neutron

### Networks

```
# List all networks
openstack network list

# List networks with more details
openstack network list --long

# Show details of a network
openstack network show private-net
```

```
# Create a tenant (private) network
openstack network create private-net

# Create a flat provider (external) network (admin only)
openstack network create \
    --external \
    --provider-physical-network physnet1 \
    --provider-network-type flat \
    external-net

# Create a VLAN provider network
openstack network create \
    --provider-physical-network physnet1 \
    --provider-network-type vlan \
    --provider-segment 100 \
    vlan100-net

# Create a VXLAN tenant network
openstack network create \
    --provider-network-type vxlan \
    --provider-segment 1001 \
    vxlan-net

# Update network (e.g., set as shared)
openstack network set --share private-net

# Delete a network
openstack network delete private-net
```

---

## Subnets

```
# List all subnets
openstack subnet list

# Show subnet details (DHCP, DNS, allocation pools, routes)
openstack subnet show private-subnet

# Create a subnet with DHCP
openstack subnet create \
    --network private-net \
    --subnet-range 192.168.100.0/24 \
    --gateway 192.168.100.1 \
    --dns-nameserver 8.8.8.8 \
```

```

--dns-nameserver 8.8.4.4 \
private-subnet

# Create an external subnet with allocation pool
openstack subnet create \
  --network external-net \
  --subnet-range 10.0.0.0/24 \
  --gateway 10.0.0.1 \
  --allocation-pool start=10.0.0.100,end=10.0.0.200 \
  --no-dhcp \
  external-subnet
# --no-dhcp: external networks usually don't need DHCP

# Add a static route to subnet
openstack subnet set \
  --host-route destination=172.16.0.0/12,gateway=192.168.100.254 \
  private-subnet

# Update DNS on existing subnet
openstack subnet set \
  --dns-nameserver 1.1.1.1 \
  private-subnet

# Delete a subnet
openstack subnet delete private-subnet

```

---

## Routers

```

# List all routers
openstack router list

# Show router details (external gateway, interfaces)
openstack router show my-router

# Create a router
openstack router create my-router

# Set external gateway on router (connects to external/provider network)
openstack router set \
  --external-gateway external-net \
  my-router
# This enables NAT (SNAT) for instances to reach the internet

# Add an internal subnet interface to router

```

```
openstack router add subnet my-router private-subnet
# This gives router an IP on the subnet so instances can use it as gateway

# Add a port to router
openstack router add port my-router <port-id>

# Remove subnet from router
openstack router remove subnet my-router private-subnet

# Remove external gateway
openstack router unset --external-gateway my-router

# Enable/disable SNAT (Source NAT) on router
openstack router set \
    --enable-snat \
    my-router

# Delete a router (must remove all interfaces first)
openstack router remove subnet my-router private-subnet
openstack router delete my-router
```

---

## Ports

```
# List all ports
openstack port list

# List ports on a specific network
openstack port list --network private-net

# List ports for a specific server
openstack port list --server my-vm

# Show port details (MAC, IP, security groups, device owner)
openstack port show <port-id>

# Create a port with a fixed IP
openstack port create \
    --network private-net \
    --fixed-ip subnet=private-subnet,ip-address=192.168.100.50 \
    reserved-port

# Create a port with no security groups (for SR-IOV / trusted VMs)
openstack port create \
    --network private-net \
```

```
--no-security-group \  
--disable-port-security \  
trusted-port  
  
# Add port to a server  
openstack server add port my-vm <port-id>  
  
# Update a port (change security group, admin state)  
openstack port set \  
  --security-group web-sg \  
  <port-id>  
  
# Disable port (blocks all traffic)  
openstack port set --disable <port-id>  
  
# Delete a port  
openstack port delete <port-id>
```

---

## Network Agents

```
# List all Neutron agents and their health  
openstack network agent list  
# Shows: Agent Type, Host, Availability Zone, Alive, State  
  
# Show details of a specific agent  
openstack network agent show <agent-id>  
  
# List DHCP agents hosting a network  
openstack network agent list \  
  --network private-net \  
  --agent-type dhcp  
  
# Schedule a network to a specific DHCP agent  
openstack network agent add network \  
  --dhcp <agent-id> private-net  
  
# Remove network from a DHCP agent  
openstack network agent remove network \  
  --dhcp <agent-id> private-net
```

---

## 5. Images – Glance

```
# List all images
openstack image list

# List all images including private and community
openstack image list --all

# Filter by status
openstack image list --status active
openstack image list --status queued

# Show image details (format, size, disk format, properties)
openstack image show ubuntu-22.04

# Upload an image from local file
openstack image create \
  --file ubuntu-22.04-cloud.img \
  --disk-format qcow2 \
  --container-format bare \
  --public \
  "Ubuntu 22.04"

# Upload a raw image
openstack image create \
  --file rhel9.raw \
  --disk-format raw \
  --container-format bare \
  --public \
  "RHEL 9"

# Download an image to local file
openstack image save \
  --file ./downloaded-image.qcow2 \
  ubuntu-22.04

# Make a private image public
openstack image set --public ubuntu-22.04

# Make an image private
openstack image set --private my-custom-image

# Set image properties (metadata)
openstack image set \
  --property hw_disk_bus=virtio \
  --property hw_vif_model=virtio \
  --property os_type=linux \
  ubuntu-22.04
```

```
# Add minimum RAM/disk requirements to image
openstack image set \
    --min-ram 1024 \
    --min-disk 10 \
    ubuntu-22.04

# Deactivate an image (prevent new instances from using it)
openstack image set --deactivate ubuntu-22.04

# Reactivate
openstack image set --activate ubuntu-22.04

# Delete an image
openstack image delete ubuntu-22.04

# Show image members (for shared images)
openstack image member list ubuntu-22.04

# Share image with another project
openstack image add project ubuntu-22.04 <target-project-id>
openstack image set --accept ubuntu-22.04 # accept on the target project side
```

---

## 6. Block Storage – Cinder

### Volumes

```
# List volumes in current project
openstack volume list

# List volumes across all projects (admin)
openstack volume list --all-projects

# Show volume details
openstack volume show my-volume

# Create a volume
openstack volume create \
    --size 100 \
    --description "Database storage" \
    db-volume

# --size in GB
```

```
# Create a volume from an image (bootable)
openstack volume create \
    --image ubuntu-22.04 \
    --size 50 \
    boot-volume
# Creates a bootable volume pre-loaded with the OS image

# Create a volume from a snapshot
openstack volume create \
    --snapshot my-snapshot \
    --size 100 \
    restored-volume

# Create a volume from another volume (clone)
openstack volume create \
    --source my-volume \
    --size 100 \
    cloned-volume

# Create a volume in a specific availability zone
openstack volume create \
    --size 50 \
    --availability-zone nova-az1 \
    az1-volume

# Attach volume to a running instance
openstack server add volume my-vm db-volume

# Attach to a specific device path
openstack server add volume \
    --device /dev/vdb \
    my-vm db-volume

# Detach volume from instance
openstack server remove volume my-vm db-volume

# Extend (grow) a volume – must be detached or instance supports online resize
openstack volume set --size 200 my-volume

# Set volume as bootable
openstack volume set --bootable my-volume

# Retype volume (change storage backend/type)
openstack volume retype \
    --migration-policy on-demand \
    my-volume ssd-type
```

```
# Delete a volume
openstack volume delete my-volume
```

---

## Volume Snapshots & Backups

```
# Create a snapshot of a volume
openstack volume snapshot create \
  --volume my-volume \
  --force \
  my-snapshot
# --force: allows snapshot of attached volume

# List snapshots
openstack volume snapshot list

# Show snapshot details
openstack volume snapshot show my-snapshot

# Delete a snapshot
openstack volume snapshot delete my-snapshot

# Create a backup (stored in Swift/S3)
openstack volume backup create \
  --name my-backup \
  --incremental \
  my-volume
# --incremental: only changed blocks since last backup

# List backups
openstack volume backup list

# Restore a backup to a volume
openstack volume backup restore \
  my-backup \
  my-restored-volume

# Delete backup
openstack volume backup delete my-backup
```

---

## Volume Types & QoS

```
# List volume types
openstack volume type list

# Show volume type details
openstack volume type show ssd-type

# Create a volume type
openstack volume type create \
  --description "High-performance SSD storage" \
  ssd-type

# Set backend-specific properties on volume type
openstack volume type set \
  --property volume_backend_name=ssd_backend \
  ssd-type

# Create QoS spec
openstack volume qos create \
  --consumer both \
  --property read_iops_sec=1000 \
  --property write_iops_sec=500 \
  ssd-qos

# Associate QoS with a volume type
openstack volume qos associate ssd-qos ssd-type

# Cinder service list (check backend health)
openstack volume service list
```

---

## 7. Object Storage – Swift

```
# List all containers
openstack container list

# Create a container
openstack container create my-bucket

# Show container details (object count, size)
openstack container show my-bucket

# Delete a container
openstack container delete my-bucket
```

```

# List objects in a container
openstack object list my-bucket

# Upload a file to a container
openstack object create my-bucket ./myfile.txt

# Upload with a different object name
openstack object create my-bucket ./myfile.txt \
    --name backups/2024/myfile.txt

# Download an object
openstack object save my-bucket myfile.txt

# Download to specific path
openstack object save \
    --filename /tmp/downloaded-file.txt \
    my-bucket myfile.txt

# Show object details (size, content-type, etag)
openstack object show my-bucket myfile.txt

# Delete an object
openstack object delete my-bucket myfile.txt

# Set object metadata
openstack object set \
    --property content-type=application/json \
    my-bucket data.json

# Create a temporary URL (time-limited download link)
openstack object store account set \
    --property Temp-URL-Key=mysecretkey

# Generate temp URL (valid for 3600 seconds)
swift tempurl GET 3600 \
    /v1/AUTH_<project-id>/my-bucket/myfile.txt \
    mysecretkey

```

---

## 8. Orchestration – Heat

```

# List all stacks
openstack stack list

# Show stack details and outputs

```

```
openstack stack show my-stack
```

```
# Create a stack from a template file
```

```
openstack stack create \  
  --template my-template.yaml \  
  --parameter key_name=mykey \  
  --parameter image=ubuntu-22.04 \  
  --parameter flavor=m1.medium \  
  my-stack
```

```
# Create with environment file
```

```
openstack stack create \  
  --template my-template.yaml \  
  --environment prod-environment.yaml \  
  my-stack
```

```
# Update a stack (change parameters or template)
```

```
openstack stack update \  
  --template my-template.yaml \  
  --parameter flavor=m1.large \  
  my-stack
```

```
# Preview changes before updating
```

```
openstack stack update \  
  --template updated-template.yaml \  
  --dry-run \  
  my-stack
```

```
# Show stack outputs (e.g., IP addresses, URLs)
```

```
openstack stack output list my-stack  
openstack stack output show my-stack server_ip
```

```
# List events (deployment history)
```

```
openstack stack event list my-stack
```

```
# Show resource details within a stack
```

```
openstack stack resource list my-stack  
openstack stack resource show my-stack my_server
```

```
# Suspend a stack (pauses all resources)
```

```
openstack stack suspend my-stack
```

```
# Resume a stack
```

```
openstack stack resume my-stack
```

```
# Abandon a stack (delete stack record but keep resources)
```

```
openstack stack abandon my-stack
```

```
# Delete a stack (destroys all resources in it)
openstack stack delete my-stack

# Validate a Heat template without deploying
openstack orchestration template validate \
  --template my-template.yaml
```

---

## 9. Placement Service

```
# List all resource providers (hypervisors, shared storage, etc.)
openstack resource provider list

# Show resources on a provider
openstack resource provider show <provider-uuid>

# Show available inventory on a resource provider
openstack resource provider inventory list <provider-uuid>
# Shows: VCPU, MEMORY_MB, DISK_GB, and their total/reserved/used

# Show current usage on a resource provider
openstack resource provider usage show <provider-uuid>

# List resource classes
openstack resource class list
# Standard: VCPU, MEMORY_MB, DISK_GB, VGPU, FPGA, NET_BW_EGR_KILOBIT_PER_SEC

# List traits (capabilities) of a resource provider
openstack resource provider trait list <provider-uuid>
# e.g., HW_CPU_X86_AVX2, COMPUTE_VOLUME_MULTI_ATTACH

# Show all allocations for an instance
openstack resource provider allocation show <instance-uuid>

# List allocation candidates for a resource request
openstack allocation candidate list \
  --resource VCPU=2 \
  --resource MEMORY_MB=4096 \
  --resource DISK_GB=40
# Shows which hypervisors can host a VM with these requirements
```

---

# 10. Security Groups & Floating IPs

## Security Groups

```
# List security groups
openstack security group list

# Show rules in a security group
openstack security group rule list my-sg
openstack security group rule list --long my-sg

# Create a security group
openstack security group create \
    --description "Rules for web servers" \
    web-sg

# Add ingress rule: allow SSH from anywhere
openstack security group rule create \
    --protocol tcp \
    --dst-port 22 \
    --remote-ip 0.0.0.0/0 \
    --ingress \
    web-sg

# Allow ICMP (ping)
openstack security group rule create \
    --protocol icmp \
    --ingress \
    web-sg

# Allow HTTP and HTTPS
openstack security group rule create --protocol tcp --dst-port 80 web-sg
openstack security group rule create --protocol tcp --dst-port 443 web-sg

# Allow a port range
openstack security group rule create \
    --protocol tcp \
    --dst-port 8000:8080 \
    web-sg

# Allow traffic from another security group (for inter-tier communication)
openstack security group rule create \
    --protocol tcp \
    --dst-port 3306 \
    --remote-group app-sg \
    db-sg
```

```
# Allows app-sg members to access port 3306 on db-sg members

# Allow all UDP
openstack security group rule create \
    --protocol udp \
    --dst-port 1:65535 \
    web-sg

# Allow all traffic from a specific CIDR
openstack security group rule create \
    --protocol any \
    --remote-ip 10.0.0.0/8 \
    internal-sg

# Add security group to a running instance
openstack server add security group my-vm web-sg

# Remove security group from instance
openstack server remove security group my-vm web-sg

# Delete a security group rule
openstack security group rule delete <rule-id>

# Delete a security group
openstack security group delete web-sg
```

---

## Floating IPs

```
# List floating IPs
openstack floating ip list

# Allocate a new floating IP from external network
openstack floating ip create external-net

# Assign floating IP to an instance
openstack server add floating ip my-vm 10.0.0.105

# Disassociate floating IP from instance (keeps it allocated)
openstack server remove floating ip my-vm 10.0.0.105

# Release floating IP back to pool
openstack floating ip delete 10.0.0.105

# Create floating IP with specific address (if available)
```

```
openstack floating ip create \  
  --floating-ip-address 10.0.0.110 \  
  external-net  
  
# Associate floating IP to a specific port  
openstack floating ip set \  
  --port <port-id> \  
  10.0.0.105
```

---

## 11. Advanced Commands

### Output Formatting

```
# Output as JSON (for scripting/piping)  
openstack server list -f json  
  
# Output as YAML  
openstack server list -f yaml  
  
# Output as CSV  
openstack server list -f csv  
  
# Show only specific columns  
openstack server list -c Name -c Status -c Networks  
  
# Output a single field value (for scripting)  
openstack server show my-vm -f value -c id  
# Returns just the UUID – useful in scripts  
  
# Format table with custom columns  
openstack server list \  
  -c ID -c Name -c Status -c "OS-EXT-SRV-ATTR:host" \  
  --long  
  
# Count resources  
openstack server list -f value | wc -l
```

---

### Quotas

```
# Show compute quotas for current project  
openstack quota show
```

```
# Show quota for a specific project
openstack quota show --project myproject

# Update compute quotas
openstack quota set \
    --instances 50 \
    --cores 200 \
    --ram 512000 \
    --floating-ips 20 \
    myproject

# Update network quotas
openstack quota set \
    --networks 20 \
    --subnets 50 \
    --ports 200 \
    --routers 10 \
    myproject

# Update storage quotas
openstack quota set \
    --volumes 100 \
    --snapshots 100 \
    --gigabytes 10000 \
    myproject

# Show quota usage details
openstack quota show --detail myproject
```

---

## Metadata & Tags

```
# Set metadata on a server
openstack server set \
    --property environment=production \
    --property owner=team-a \
    my-vm

# Remove a metadata property
openstack server unset --property environment my-vm

# Add tags to server
openstack server set \
    --tag web \
```

```
--tag production \  
my-vm  
  
# Filter servers by tag  
openstack server list --tags web,production  
  
# Set metadata on a volume  
openstack volume set \  
    --property backup=daily \  
my-volume  
  
# Set metadata on an image  
openstack image set \  
    --property os_distro=ubuntu \  
    --property os_version=22.04 \  
ubuntu-22.04
```

---

## Aggregates & Host Zones

```
# List host aggregates  
openstack aggregate list  
  
# Show aggregate details  
openstack aggregate show gpu-aggregate  
  
# Create a host aggregate  
openstack aggregate create \  
    --zone gpu-zone \  
gpu-aggregate  
  
# Add a compute host to aggregate  
openstack aggregate add host gpu-aggregate gpu-compute01  
  
# Remove a host from aggregate  
openstack aggregate remove host gpu-aggregate gpu-compute01  
  
# Set metadata on aggregate (used for flavor scheduling)  
openstack aggregate set \  
    --property gpu=true \  
gpu-aggregate  
  
# Then on the flavor, require this property:  
openstack flavor set \  
    --property aggregate_instance_extra_specs:gpu=true \  

```

```
gpu-flavor
# Instances using gpu-flavor will ONLY land on hosts in gpu-aggregate
```

---

## Barbican (Key Management)

```
# Store a secret
openstack secret store \
  --name my-ssl-cert \
  --payload "$(cat my-cert.pem)" \
  --payload-content-type text/plain

# List secrets
openstack secret list

# Get a secret value
openstack secret get --payload \
  https://barbican:9311/v1/secrets/<secret-id>

# Delete a secret
openstack secret delete <secret-id>
```

---

## Ceilometer / Gnocchi (Telemetry)

```
# List available metrics
openstack metric list

# Show metric details
openstack metric show <metric-id>

# Query resource metrics (e.g., CPU utilization for an instance)
openstack metric measures show \
  --resource-id <instance-id> \
  --aggregation mean \
  cpu_util

# List alarms
openstack alarm list

# Create a CPU alarm
openstack alarm create \
  --type gnocchi_resources_threshold \
```

```
--name high-cpu-alarm \  
--metric cpu_util \  
--threshold 80.0 \  
--comparison-operator gt \  
--aggregation-method mean \  
--granularity 300 \  
--resource-type instance \  
--resource-id <instance-id> \  
--alarm-action http://webhook.example.com/alert
```

---

## Load Balancer – Octavia

```
# Create a load balancer  
openstack loadbalancer create \  
  --name my-lb \  
  --vip-subnet-id private-subnet  
  
# Wait for LB to become active  
openstack loadbalancer show my-lb  
  
# Create a listener (frontend)  
openstack loadbalancer listener create \  
  --name my-listener \  
  --protocol HTTP \  
  --protocol-port 80 \  
  my-lb  
  
# Create a pool (backend group)  
openstack loadbalancer pool create \  
  --name my-pool \  
  --lb-algorithm ROUND_ROBIN \  
  --listener my-listener \  
  --protocol HTTP  
  
# Add members to pool  
openstack loadbalancer member create \  
  --subnet-id private-subnet \  
  --address 192.168.100.10 \  
  --protocol-port 8080 \  
  my-pool  
  
openstack loadbalancer member create \  
  --subnet-id private-subnet \  
  --address 192.168.100.11 \
```

```
--protocol-port 8080 \  
my-pool  
  
# Create health monitor  
openstack loadbalancer healthmonitor create \  
  --delay 5 \  
  --max-retries 3 \  
  --timeout 5 \  
  --type HTTP \  
  --url-path /health \  
  my-pool  
  
# List load balancers  
openstack loadbalancer list  
  
# Show LB status tree  
openstack loadbalancer status show my-lb  
  
# Delete load balancer  
openstack loadbalancer delete --cascade my-lb
```

---

## DNS – Designate

```
# List DNS zones  
openstack zone list  
  
# Create a DNS zone  
openstack zone create \  
  --email admin@example.com \  
  example.com.  
  
# Create a DNS record  
openstack recordset create \  
  --records 10.0.0.50 \  
  --type A \  
  example.com. \  
  www  
  
# List records in a zone  
openstack recordset list example.com.  
  
# Delete a record  
openstack recordset delete example.com. www
```

```
# Delete a zone
openstack zone delete example.com.
```

---

## 12. Troubleshooting Commands

### Instance Troubleshooting

```
# Check instance status and fault message
openstack server show my-vm | grep -E "status|fault|OS-EXT"
# Look for: fault, task_state, vm_state, power_state

# Full instance details including scheduling info
openstack server show my-vm -f json | python3 -m json.tool

# Check which compute host an instance is on
openstack server show my-vm \
  -c "OS-EXT-SRV-ATTR:host" \
  -c "OS-EXT-SRV-ATTR:hypervisor_hostname" \
  -c "OS-EXT-SRV-ATTR:instance_name"

# Force instance out of ERROR state back to ACTIVE
openstack server reset-state --active my-vm
# WARNING: Use only if hypervisor is actually running the instance

# Force instance into ERROR state (to allow deletion)
openstack server reset-state my-vm

# View last 100 lines of instance console log
openstack console log show --lines 100 my-vm

# List all actions performed on an instance (history)
openstack server event list my-vm

# Show details of a specific action (e.g., why a resize failed)
openstack server event show my-vm <request-id>

# Check if instance is reachable from OpenStack perspective
openstack server list --status ACTIVE | grep my-vm

# List instance volumes (check if all volumes are attached)
openstack volume list --server my-vm
```

---

## Network Troubleshooting

```
# Check all Neutron agents are up
openstack network agent list
# Look for Alive=True and Admin State Up=True for all agents

# Find which DHCP agent serves a network
openstack network agent list \
    --network private-net \
    --agent-type dhcp

# Check DHCP namespace on the network node (run on network node)
ip netns list | grep qdhcp
ip netns exec qdhcp-<network-id> ip addr
ip netns exec qdhcp-<network-id> cat /var/lib/neutron/dhcp/<net-id>/leases

# Check router namespace
ip netns list | grep qrouter
ip netns exec qrouter-<router-id> ip route
ip netns exec qrouter-<router-id> ping 8.8.8.8

# Check router external gateway is set
openstack router show my-router | grep external_gateway_info

# List all floating IPs and their associations
openstack floating ip list

# Check if a port exists and has IP
openstack port list --server my-vm

# Find why instance has no IP (check port status)
openstack port list --server my-vm -c ID -c "Fixed IP Addresses" -c Status
# Status should be ACTIVE, not DOWN or BUILD

# Check OVS bridge config (on compute/network node)
ovs-vsctl show
ovs-vsctl list-br
ovs-ofctl show br-int

# Check OVN (if using OVN backend)
ovn-nbctl show
ovn-sbctl show
ovn-nbctl ls-list      # logical switches
ovn-nbctl lr-list      # logical routers
```

---

## Volume Troubleshooting

```
# Check Cinder services health
openstack volume service list
# Look for: State=up and Status=enabled

# Show volume details including error message
openstack volume show my-volume | grep -E "status|error|attach|migration"

# List volumes in error state
openstack volume list --status error

# Reset volume state (e.g., from error to available)
openstack volume set --state available my-volume
# WARNING: Only do this if the volume is actually not in use

# Check volume attachments
openstack volume attachment list
openstack volume attachment show <attachment-id>

# Force delete a stuck volume
openstack volume delete --force my-volume

# Check which backend a volume is on
openstack volume show my-volume | grep "os-vol-host-attr:host"

# List volume transfers (between projects)
openstack volume transfer request list
```

---

## Authentication / Keystone Troubleshooting

```
# Test if Keystone is responding
curl -s http://192.168.1.10:5000/v3/ | python3 -m json.tool

# Issue a token manually (test credentials)
openstack token issue

# Check if token is valid
openstack token issue -f value -c id

# Check Keystone service is in catalog
openstack catalog list
openstack catalog show identity
```

```
# List Keystone endpoints and check URLs
openstack endpoint list --service identity

# Check if service account exists
openstack user show nova
openstack user show glance
openstack user show neutron

# Verify service user has admin role on service project
openstack role assignment list \
  --user nova \
  --project service \
  --names
```

---

## Quota & Resource Troubleshooting

```
# Check if quota is exhausted
openstack quota show

# Compare "used" vs "limit" for each resource type

# Check Nova quota usage in detail
openstack limits show --absolute
openstack limits show --rate

# Find unused floating IPs (allocated but not associated)
openstack floating ip list | grep "None"

# Find stale ports (no device attached)
openstack port list | grep ""

# Find orphaned volumes (available but old)
openstack volume list --status available

# Show per-project resource usage (admin only)
openstack usage list
openstack usage show --project myproject
```

---

## 13. Service-Level Troubleshooting

### Nova (Compute)

```
# Check nova services
openstack compute service list

# Check nova-api is working
nova service-list    # legacy nova CLI

# Check placement is reachable from nova
nova-manage cell_v2 list_cells
nova-manage cell_v2 verify_host --host compute01.example.com

# List hosts in a cell
nova-manage cell_v2 list_hosts

# Discover new compute hosts
nova-manage cell_v2 discover_hosts --verbose

# Check nova DB migrations
nova-manage db version
nova-manage api_db version

# Analyze scheduler (why instance didn't land on a host)
openstack server show <failed-vm> -c fault
# Then check: nova-scheduler logs for "NoValidHost"

# Force evacuate from failed compute host
nova-manage compute delete --updated-since <timestamp>
```

---

## Neutron (Networking)

```
# Check Neutron server log for errors
sudo journalctl -u neutron-server -n 100 --no-pager
sudo journalctl -u neutron-server -f    # follow live

# Check OVS agent logs
sudo journalctl -u neutron-openvswitch-agent -n 100

# Check L3 agent log
sudo journalctl -u neutron-l3-agent -n 100

# Check DHCP agent log
sudo journalctl -u neutron-dhcp-agent -n 100

# Debug connectivity from router namespace
```

```
ROUTER_NS=$(ip netns list | grep qrouter | head -1 | awk '{print $1}')
sudo ip netns exec $ROUTER_NS ping -c3 8.8.8.8
sudo ip netns exec $ROUTER_NS ip route
sudo ip netns exec $ROUTER_NS iptables -t nat -L -n -v | head -30

# Debug DHCP from DHCP namespace
NET_ID=$(openstack network show private-net -c id -f value)
sudo ip netns exec qdhcp-${NET_ID} netstat -nlup | grep dnsmasq
sudo ip netns exec qdhcp-${NET_ID} cat /var/lib/neutron/dhcp/${NET_ID}/leases
```

---

## Glance (Image Service)

```
# Check Glance API status
openstack image list

# Check Glance service log
sudo journalctl -u glance-api -n 100

# Check image store path
ls -la /var/lib/glance/images/

# Check image is not corrupted (check size matches)
openstack image show ubuntu-22.04 | grep size

# Re-upload a corrupted image
openstack image set --deactivate ubuntu-22.04
openstack image delete ubuntu-22.04
# Then re-create
```

---

## Cinder (Block Storage)

```
# Check Cinder services
openstack volume service list

# Check c-vol log
sudo journalctl -u cinder-volume -n 100

# Check c-api log
sudo journalctl -u cinder-api -n 100

# Check c-scheduler log
```

```
sudo journalctl -u cinder-scheduler -n 100
```

```
# LVM backend: check volume group
```

```
sudo vgdisplay cinder-volumes
```

```
sudo lvdisplay | grep cinder
```

```
# Reset stuck migration
```

```
openstack volume set --state available my-volume
```

```
# Sync volume quotas (fixes usage drift)
```

```
cinder-manage quota sync --project-id <project-id>
```

---

## 14. Log Locations Reference

| Service            | Log Path (default)                                      |
|--------------------|---------------------------------------------------------|
| Keystone           | /var/log/keystone/keystone.log or journalctl -u apache2 |
| Nova API           | /var/log/nova/nova-api.log                              |
| Nova Compute       | /var/log/nova/nova-compute.log                          |
| Nova Scheduler     | /var/log/nova/nova-scheduler.log                        |
| Nova Conductor     | /var/log/nova/nova-conductor.log                        |
| Glance             | /var/log/glance/api.log                                 |
| Neutron Server     | /var/log/neutron/neutron-server.log                     |
| Neutron OVS Agent  | /var/log/neutron/openvswitch-agent.log                  |
| Neutron L3 Agent   | /var/log/neutron/l3-agent.log                           |
| Neutron DHCP Agent | /var/log/neutron/dhcp-agent.log                         |
| Cinder API         | /var/log/cinder/cinder-api.log                          |
| Cinder Volume      | /var/log/cinder/cinder-volume.log                       |
| Cinder Scheduler   | /var/log/cinder/cinder-scheduler.log                    |
| Heat API           | /var/log/heat/heat-api.log                              |
|                    |                                                         |

```
# Systemd journal (all-in-one log search)
sudo journalctl -u nova-compute --since "1 hour ago"
sudo journalctl -u neutron-server --grep "ERROR"
sudo journalctl -u cinder-volume -p err      # only errors and above

# Tail multiple logs simultaneously
sudo tail -f /var/log/nova/nova-api.log \
            /var/log/neutron/neutron-server.log \
            /var/log/cinder/cinder-api.log

# Search for a specific request ID across all logs
grep "req-XXXX-XXXX" /var/log/nova/*.log /var/log/neutron/*.log
```

---

## 15. Quick Reference Cheat Sheet

### Most-Used Commands

```
# — AUTH —————
source ~/admin-openrc.sh          # Load credentials
openstack token issue             # Verify auth works

# — INSTANCES —————
openstack server list             # List VMs
openstack server list --all-projects # All VMs (admin)
openstack server show <name/id>   # VM details
openstack server create ...       # Launch VM
openstack server stop/start <name> # Power off/on
openstack server reboot --hard <name> # Hard reboot
openstack server delete <name>    # Delete VM
openstack console log show <name>  # Boot console log
openstack console url show <name>  # VNC URL

# — NETWORK —————
openstack network list           # Networks
openstack subnet list            # Subnets
openstack router list            # Routers
openstack floating ip list       # Floating IPs
openstack floating ip create ext-net # Allocate FIP
openstack server add floating ip <vm> <fip>
openstack network agent list     # Agent health
```

```

# — IMAGES —————
openstack image list                # Images
openstack image create ...          # Upload image
openstack image show <name>        # Image details

# — VOLUMES —————
openstack volume list              # Volumes
openstack volume create --size 50 <name>
openstack server add volume <vm> <vol>
openstack volume snapshot create ... # Snapshot
openstack volume service list      # Backend health

# — IDENTITY —————
openstack user list                # Users
openstack project list            # Projects
openstack role assignment list     # Role assignments
openstack service list            # Service catalog
openstack endpoint list           # API endpoints

# — TROUBLESHOOT —————
openstack compute service list     # Nova health
openstack volume service list      # Cinder health
openstack network agent list       # Neutron health
openstack server show <vm> | grep fault # Error message
openstack server reset-state --active <vm> # Fix ERROR state
openstack volume set --state available <vol> # Fix stuck vol
openstack quota show               # Check quota limits

```

---

### Tips:

- Add `--debug` to any command for full HTTP request/response output: `openstack --debug server list`
  - Add `--os-cloud <cloud-name>` to switch clouds defined in `~/.config/openstack/clouds.yaml`
  - Use `openstack help <command>` for full option reference: `openstack help server create`
  - Request IDs in API responses (e.g., `req-xxxxx`) can be traced across all service logs
-

